Challenges and Limitations in Data Collection: learnings from the DTI project

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DTI database

Digital Trade Integration Project

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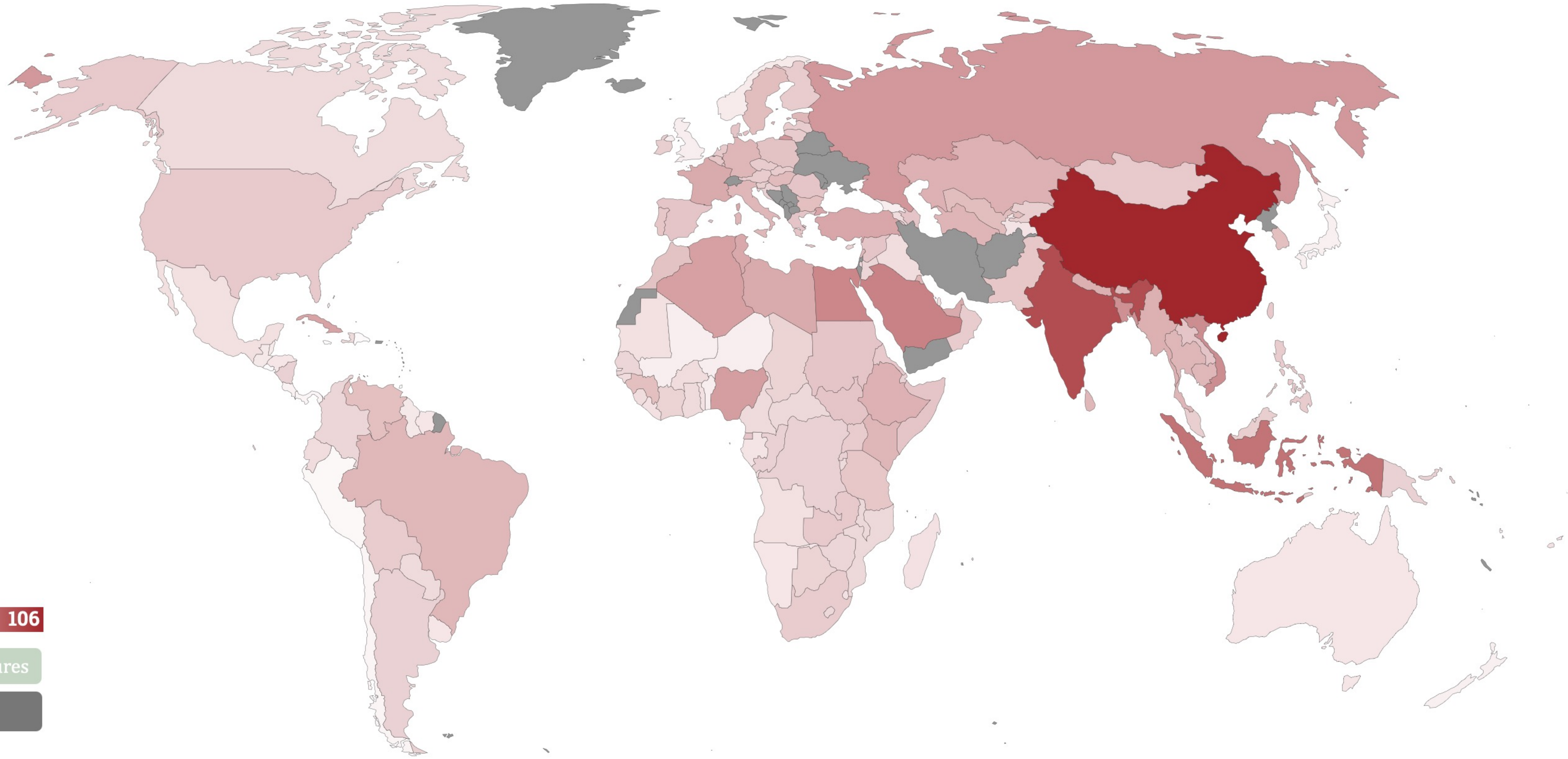
Digital trade policies and practices across the globe

The map provides a snapshot of the variety of information contained in the DTI database, which currently includes over 5000 unique measures expected to restrict digital trade integration and 1600 measures expected to enable integration. You can select one of the 12 pillars and 65 indicators for a detailed overview.

All pillars

All indicators



No. of restrictions 16 106

Restrictions Enabling measures

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Steps in Data Collection: Recruitment

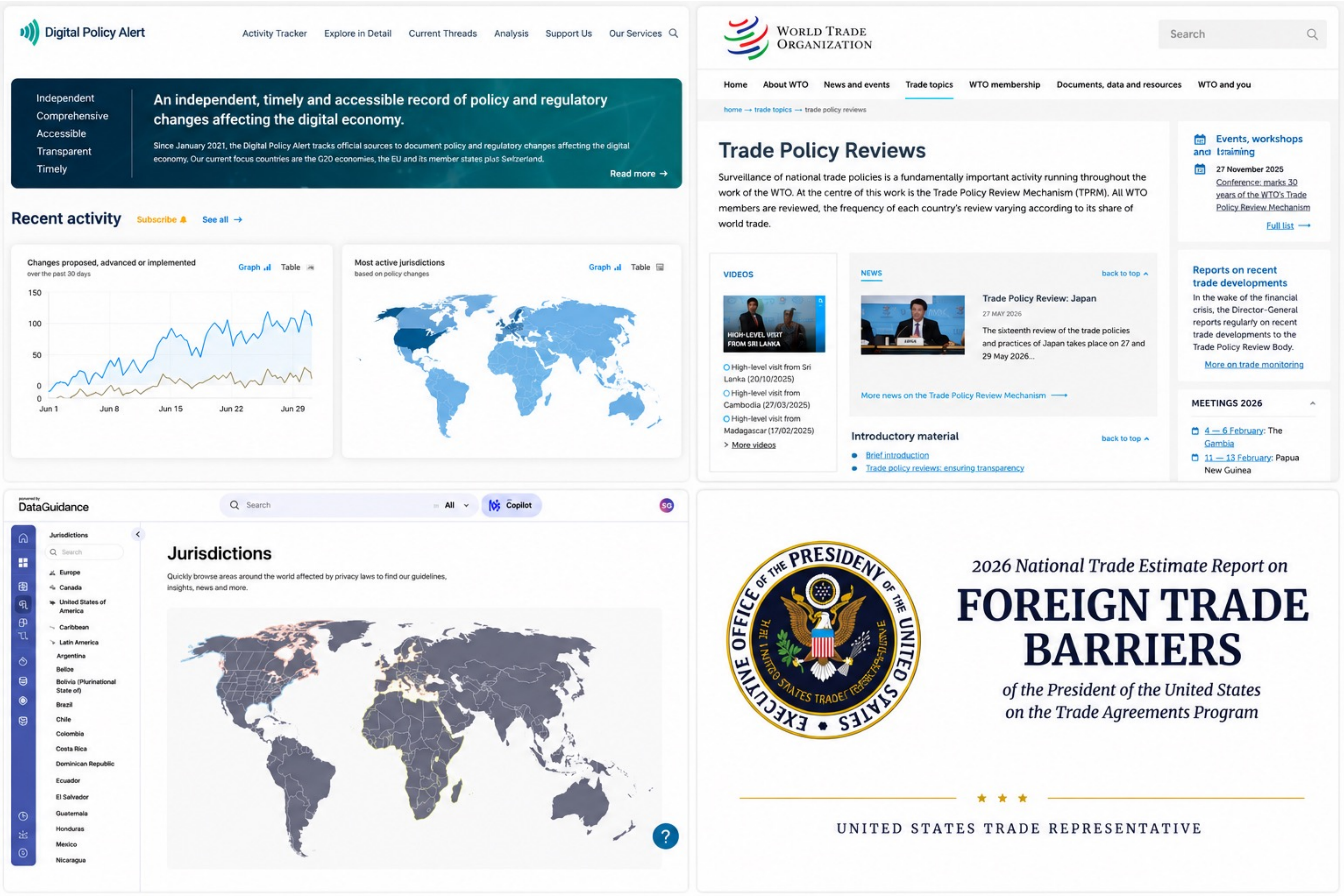
- Initial selection of practitioners and researchers from academic institutions and international organisations.
- Preferably based in the country under analysis.
- **Challenges:** recruiting and training local researchers with expertise in the local language, legal framework, and practices.
- **Risk:** balancing local knowledge and prior data collection experience.

Partners

EUI European University Institute
LSE The London School of Economics and Political Science
Teesside University
Hertie School
Università Commerciale Luigi Bocconi
United Nations Economic Commission for Africa
ECIPE European Centre for International Political Economy
United Nations ESCAP Economic and Social Commission for Asia and the Pacific
Naciones Unidas United Nations CEPAL ECLAC
Digital Cooperation Organization
ADB Asian Development Bank
ESCWA
Jean Monnet Network Trade & Investment in Services Associates (TIISA)
Co-funded by the Erasmus+ Programme of the European Union

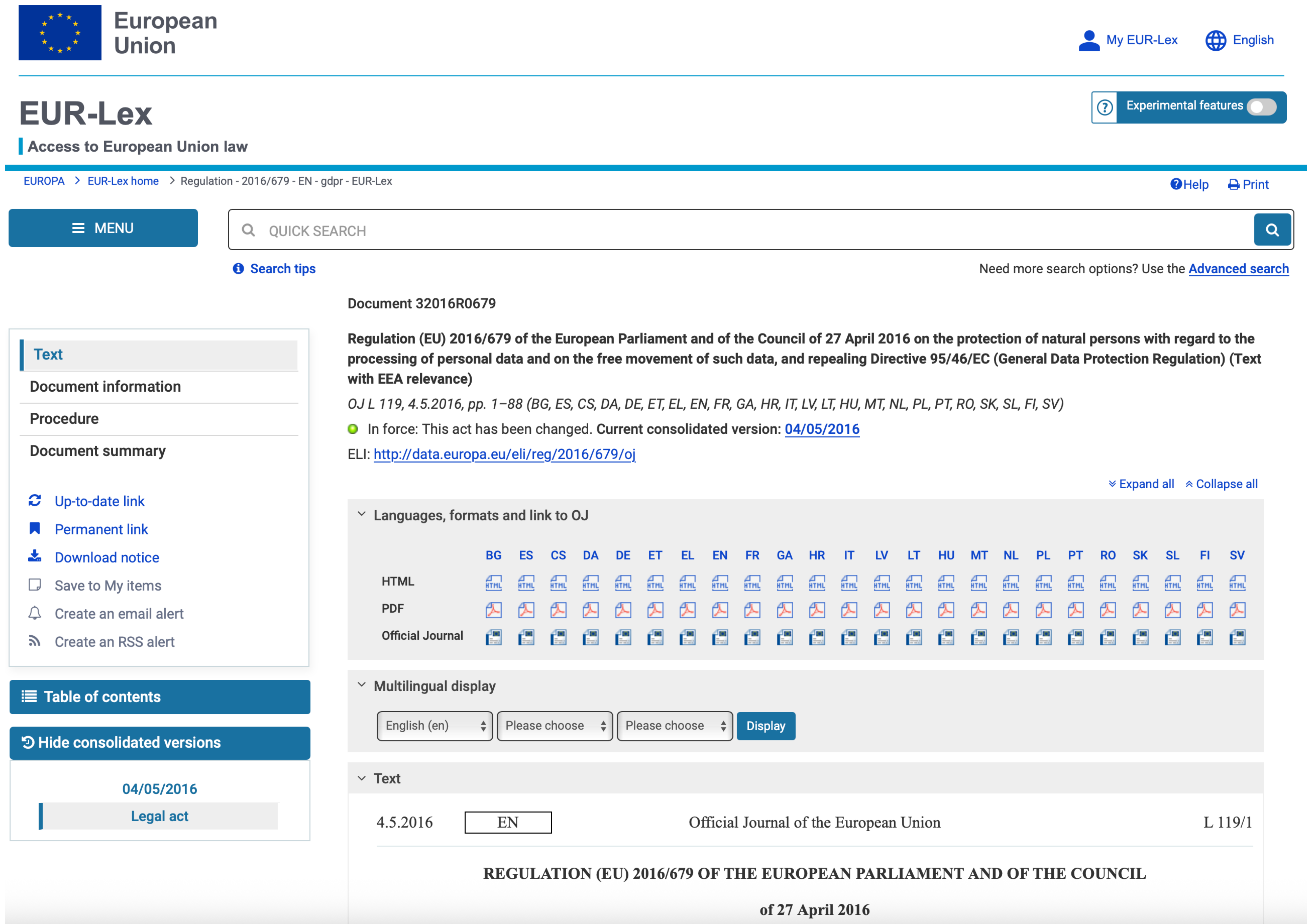
Steps in Data Collection: Review of Secondary Sources

- Including regulatory databases, reports by international organisations, legal reviews, government resources, and news articles.
- **Challenges:** limited focus on specific policy areas, lack of structured formats for efficient review, and gaps in coverage across many countries.
- **Risks:** They may oversimplify issues, omit relevant information, contain errors, or reflect partial or biased positions.



Steps in Data Collection: Review of Primary Sources

- Namely official gazettes of laws and regulations.
- **Challenges:** limited availability of legal texts in certain countries, language barriers, broken links, and difficulties in identifying relevant provisions within legislation.
- **Risks:** They may contradict or complement one another, and these relationships may not be identified or taken into account.



Time-Consuming Aspects of the Process

CHINA

Regulatory restriction

Since November 2016, entry into force in June 2017, last amended in October 2025
Since March 2024

Pillar Cross-border data policies | **Indicator** Ban to transfer and local processing requirement i

Cybersecurity Law of the People's Republic of China (中华人民共和国网络安全法)

Provisions on Promoting and Regulating the Cross-Border Flow of Data (促进和规范数据跨境流动规定)

Art. 39 of the Cybersecurity Law requires that personal information and important data collected or generated by operators of critical information infrastructure within mainland China be stored domestically, and allows their transfer abroad only when genuinely necessary for business purposes and subject to a security assessment conducted in accordance with measures issued jointly by the national cybersecurity and informatization authority and relevant State Council departments, unless other laws provide otherwise. Art. 33 defines critical information infrastructure to include sectors such as public communications and information services, energy, transport, water, finance, public services, and e-government, as well as any infrastructure whose damage, loss of function, or data leakage could seriously endanger national security, public welfare, or the public interest. Art. 7 of the Provisions on Promoting and Regulating the Cross-Border Flow of Data requires data handlers to apply for a data export security assessment when critical information infrastructure operators export personal information or important data, but Art. 5 exempts certain transfers of personal information, though not important data, where the transfer is genuinely necessary for the performance of a contract, for lawful cross-border human resources management, or for emergency protection of life, health, or property. Art. 10 obliges data handlers exporting personal information to provide notice, obtain separate consent, and conduct a personal information protection impact assessment, and Art. 11 further requires them to fulfil data security obligations and adopt technical and other necessary measures to ensure the security of exported data.

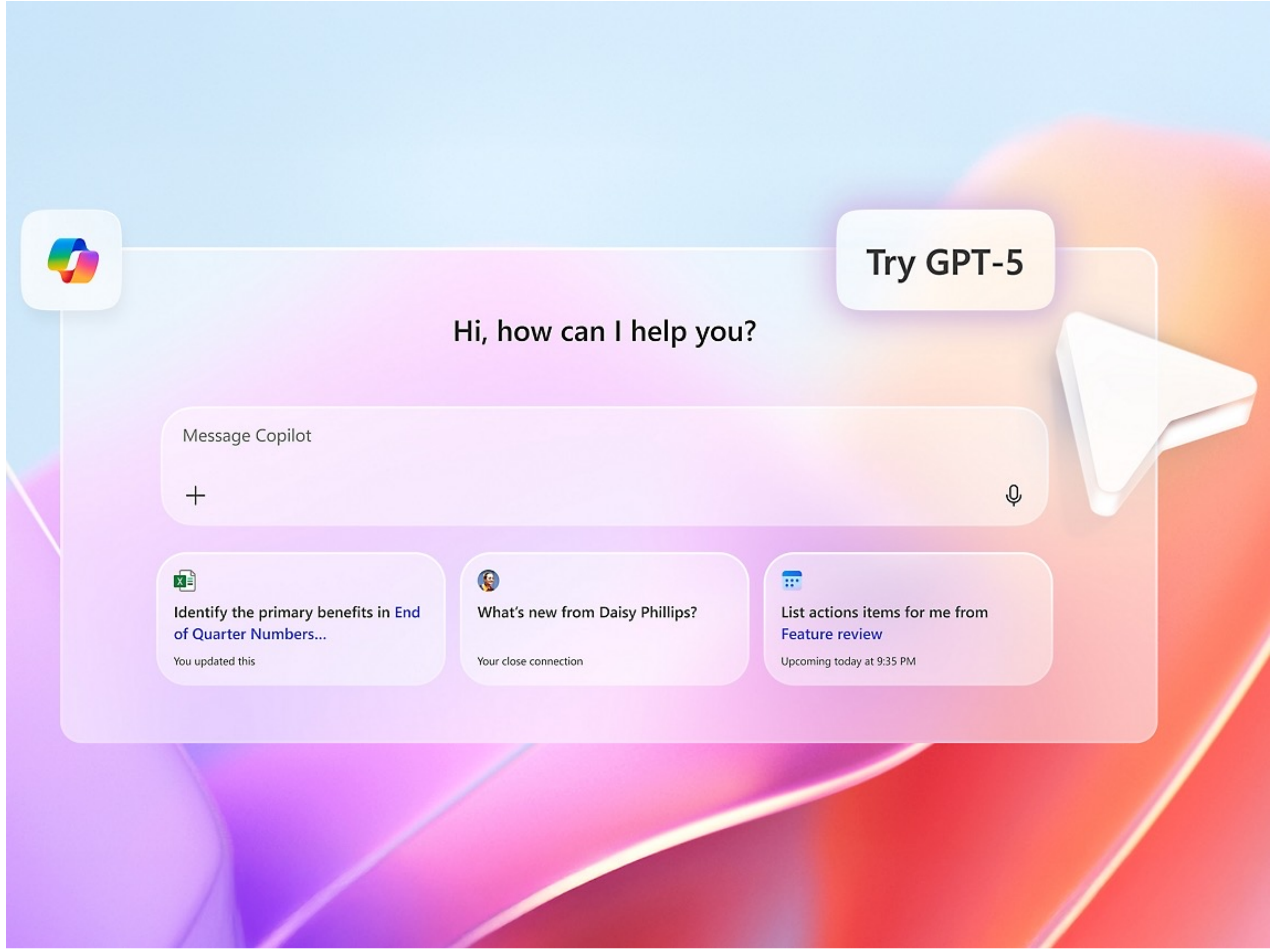
Coverage Critical information infrastructure operators

Sources

- https://web.archive.org/web/20260323221151/https://www.cac.gov.cn/2025-12/29/c_1768735112911946.htm
- https://web.archive.org/web/20260308155323/https://www.cac.gov.cn/2024-03/22/c_1712776611775634.htm

- Establishing key timelines for policies, including adoption, entry into force, amendments, and repeal.
- Ensuring accurate translation.
- Loss of access to previously verified links.
- Cross-checking by multiple reviewers to minimise errors.

Role of Tools in the Data Collection Process



- Tools for identifying relevant information in primary and secondary sources are crucial.
- The limited readability of many documents, including scanned PDFs, requires the use of specialised tools.
- Accurate translation tools are essential, as linguistic errors can significantly compromise data quality.
- Tools for ensuring link preservation are also essential.
- **Limitations:** Availability of tools to support the work, and knowledge of how to use them effectively.
- **Risk:** Automated tools cannot fully replace human verification, potentially leading to divergent interpretations.

Additional Challenges and Limitations

- Exponential growth in regulatory activity, and frequent amendments and replacement of legislation across countries.
  - Increasing technicality and complexity of regulatory policies.

- Limited responsiveness of governments to feedback requests.
  - Regulatory heterogeneity across countries and limited consensus on classification methodologies across jurisdictions.

- Lack of information online in least developed economies.

What should the tool produce?

From legal information to structured legal findings

- A useful tool should not only retrieve documents or summarise laws.

- It should **produce structured fin**dings that can be checked by researchers and integrated into a common methodology.

- Each finding should identify the rule, legal source, exact provision, original text, translation, timeline, and relevance for a specific indicator.

- The tool should **accelerate research, while leaving final classification and validation** to human reviewers.

Definition

A documented record linking a regulatory measure to its legal source, provision, text, translation, timeline, and methodological relevance.

Format: database-ready and reviewable output

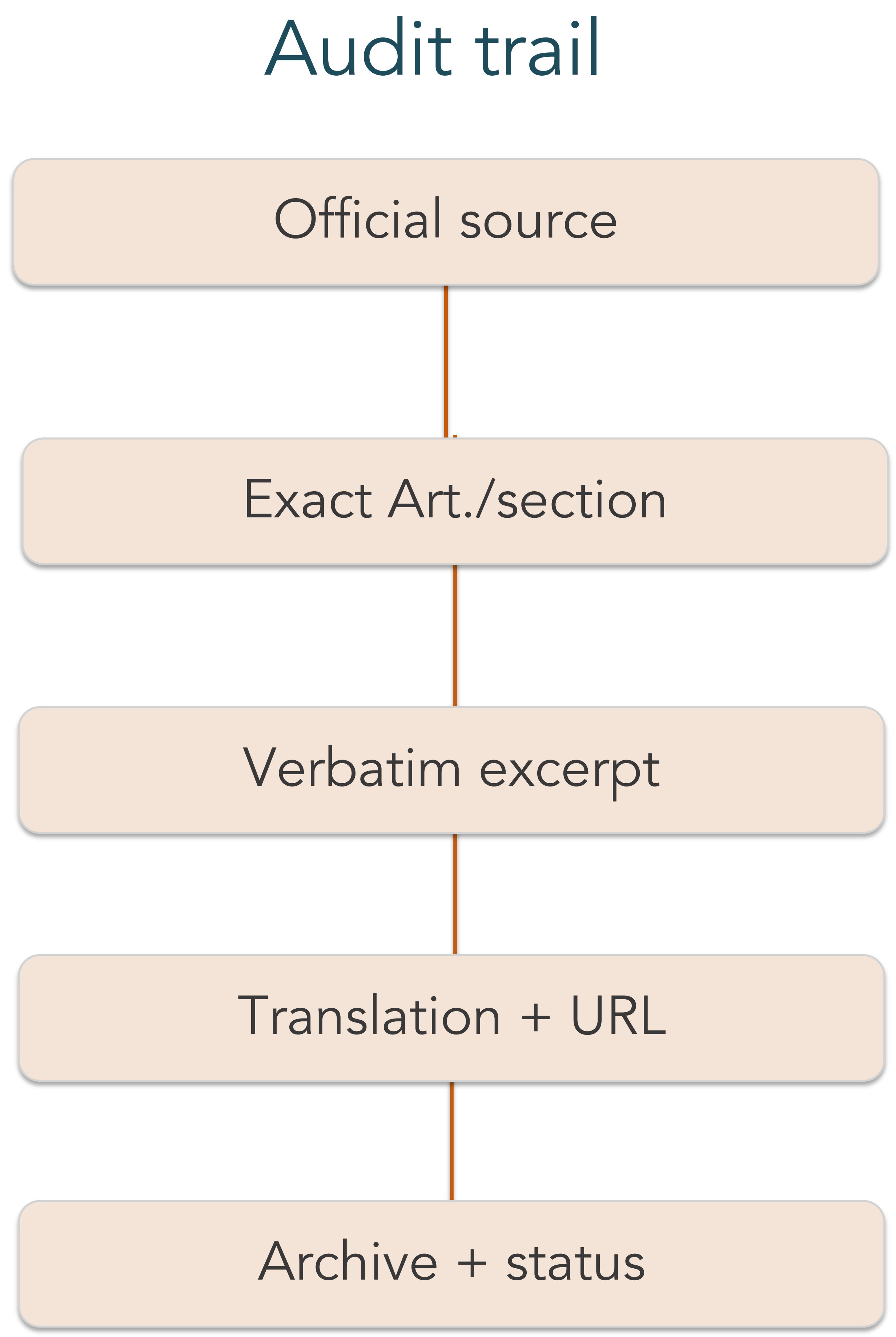
- Outputs should be machine-readable, preferably CSV, Excel, JSON or API-ready.
- Each finding should be separated into fields, not presented only as narrative text.
- The format should distinguish between source discovery, legal extraction, classification, and reviewer validation.
- It should include **flags for uncertainty**, missing information, contradictory sources, and possible amendments or repeals.

Minimum fields

Country	Pillar / indicator
Measure type	Legal instrument
Provision	Original text
Translation	Status
Timeline	Source
Confidence flags	Reviewer decision

Citation quality: evidence must be auditable

- Citations should **point to the exact legal** basis, not only to the name of the law (secondary sources are good proxies to start).
- Minimum standard: official source, provision number, quoted text, translation, URL, access date, and archived copy.
- The tool should **distinguish** primary sources from secondary sources.
- Secondary sources may guide discovery, but primary sources should support classification.
- The output should allow reviewers to verify the evidence without repeating the full search.



Language coverage: multilingual by design

- The tool should work with original-language legal texts, not only English-language sources.
- It should provide **both the original quote and an English translation**.
- Translation should **preserve legal meaning**, defined terms, institutional names, and article references.
- It should **flag uncertain translations** and terms with no direct equivalent.
- Multilingual capacity is essential for coverage across developing economies and non-English legal systems.

Original legal text — **English translation** — Uncertainty flags

Key principle

The translation should support legal interpretation, not replace the original source.

Human-in-the-loop: acceleration, not replacement

- The tool should not present uncertain classifications as final answers.
- It should not hide the reasoning path between source and classification.
- It should not remove the need for legal and methodological review.
- The best tool is an evidence-generation and verification assistant.

Bottom line

AI adds value when it reduces time spent on discovery, extraction, translation and citation checking, while keeping researchers responsible for final validation. However, all of these must me treated as **sources** and **not final answers**

THANK YOU!

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